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A multimodal learning analytics model based on deep learning for predicting student performance using tabular and time-series data fusion

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Abstract

The rapid growth of digital learning has increased interest in using multimodal learning analytics (MLA) to better understand student behavior and support more personalized teaching. This study presents a deep learning based MLA model that leverages different types of educational data such as demographics, assessment scores, online activity, and learning style information through separate feature extraction branches and an intermediate data fusion method. Our model combines Multi-Layer Perceptron (MLP) and Long Short-Term Memory (LSTM) neural networks to automatically extract and learn features from both tabular and temporal data. Furthermore, an Aggregation block is also introduced to convert sparse intermediate features into dense and more useful representations, helping the model capture stronger relationships across modalities. The proposed approach enables early identification of student performance trends, thereby supporting more informed and individualized timely interventions. Our experiment on the Open University Learning Analytics Dataset (OULAD) and a self-collected dataset show that the proposed model archives promising results. On OULAD, it reaches an average Mean Squared Error (MSE) of 0.2404, Mean Absolute Error (MAE) of 0.3413 and 0.7836 R^2 -score value. Moreover, it also achieved competitive results on our self-collected dataset where it reaches 0.7410, 0.6636 and 0.2893 for average MSE, MAE and R^2 -score value respectively. These results demonstrate the value of combining multiple data sources and the benefits of the proposed architecture. Overall, this work offers a reliable and scalable approach to learning analytics that can help create more adaptive and effective digital learning environments.

Keywords Educational data mining, Multimodal learning analytics, Data fusion, Deep learning, OULAD



1 Introduction

In recent years, digital technologies have profoundly reshaped modern education through the growth of online learning platforms, virtual classrooms, and applications of artificial intelligence (AI) aimed at optimizing teaching and learning processes [1, 2]. These innovations not only provide students with greater flexibility in how they learn but also enhance educational quality and reduce associated costs [3]. A growing body of research has sought to leverage the vast data generated by digital learning environments to unlock the full potential of data driven education. This data encompasses online interactions between students and instructors, structural elements of learning materials, curriculum design, and academic performance metrics [4].

Within the field of educational research, Educational Data Mining (EDM) and Learning Analytics (LA) have emerged as prominent paradigms for analyzing learner behaviors, diagnosing difficulties, and designing effective pedagogical interventions [4]. EDM applies data mining techniques to uncover actionable insights from educational datasets, whereas LA emphasizes the measurement, collection, analysis, and reporting of data related to learners and their environments to improve learning outcomes [5]. However, conventional EDM and LA methodologies often depend on single-source data, which limits their ability to capture the complicated and dynamic nature of the learning process. Nowadays, online learning platforms are increasingly capable of capturing a huge quantity of data from diverse sources such as demographics, behavior logs, and learning preferences. Research from the educational and behavioral sciences suggests that these multimodal factors each provide unique insights into student performance [6, 7]. Demographic attributes including age, gender, and socioeconomic status can shape a learner's motivational beliefs, which in turn influence engagement and achievement [7]. Similarly, learning preferences reflect cognitive processing tendencies and self-learning strategies, highlighting how students choose and interact with instructional materials [8]. Finally, detailed logs of weekly online interactions such as login frequency, quiz attempts or forum participation can capture students' active engagement, social presence and study routines [8]. By integrating these multimodal data, educators and researchers can better understand not only student's knowledge, but also how they learn, engage, and sustain motivation throughout the course, enabling a more theoretically grounded view of learning outcomes.

Leveraging these strengths, Multimodal Learning Analytics or MLA in short has been proposed as an approach that integrates and analyzes data from multiple sources including learning styles, assessment scores, demographic profiles, and patterns of online interaction [9]. MLA facilitates a more holistic understanding of student learning behavior and supports predictive applications such as early warning systems for identifying at-risk students [10] as well as recommendation engines that tailor learning materials to individual proficiency levels and interests [11]. Early MLA research primarily focused on modeling individual learning preferences. Classification models such as VARK [12] and the Felder-Silverman Learning Style Model [13] were widely used to categorize learners based on sensory modalities or cognitive preferences. While these models tend to oversimplify the complexity of real-world learning behavior [14], they served as valuable foundations for the development of personalized learning environments [15].

Despite recent advances, several critical limitations persist in current MLA methodologies. For example, Chango et al. [16] employed traditional machine learning techniques

for multimodal data fusion, but their models were constrained by delayed predictions and limited sample sizes. Deep learning models, such as recurrent neural networks (RNNs) including LSTM and GRU were later explored by Al-Azazi et al. [17] for sequential data analysis, however, their fusion techniques lacked optimization and resulting in unstable model performance. More recent studies, such as those by Al-Zawqari et al. [18] and Shou et al. [19] introduced hybrid architectures (e.g., LSTM-ANN, Multi-head Attention) to improve prediction accuracy, yet issues such as overfitting and inefficient training procedures remain unresolved. Furthermore, existing models often lack robustness across diverse learning contexts. For instance, Prabowo et al. [20] proposed a model combining MLP and LSTM to predict student GPA using multimodal data. However, the reliance on historical academic performance can introduce bias and limit the model's ability to adapt to behavioral shifts or detect anomalies such as academic dishonesty.

Motivated by these prior works, our approach introduces a series of architectural refinements designed to resolve the fundamental limitations of joint modality processing seen in the baseline model proposed by Prabowo et al. [20]. While previous methods often suffer from feature interference where noise from temporal data can degrade the learning of stable static attributes, our architecture implements a modal decoupling strategy [21]. By assigning each data source to a dedicated Feature Extraction block, each modality is represented and learned within its own isolated gradient path. This architecture prevents errors in one data source from affecting the others. For example, noisy or missing weekly logs will not affect the reliable patterns learned from student demographics, leading to more robust predictions [22]. Beyond simple isolation, our model moves past the shallow feature stacking used in MLP-LSTM [20] by incorporating a specialized Aggregation block. This component acts as a non-linear bottleneck that compresses sparse, high dimensional features. By doing so, it transforms them into a dense, unified representation that is easier for the model to process. Unlike the simple concatenation used in the baseline, this block enables the network to capture complex correlations between modals. It captures the complex relationship between fixed student attributes and their evolving behavioral patterns, which simple linear fusion methods often fail to detect [23]. Finally, we adopt an intermediate fusion strategy that integrates modalities only after independent refinement, thereby preserving the unique signal of each data stream to achieve greater predictive stability than early or late fusion methods [24].

In summary, we propose a multimodal learning analytics framework that enhances predictive accuracy and robustness by addressing the architectural limitations of joint modality processing found in prior works. By employing independent Feature Extraction branches, our design implements a modal decoupling strategy that prevents noise or missing values in one data stream from compromising the learning of others. This is further strengthened by an Aggregation block and an Intermediate Fusion strategy, which allow the model to capture deep, non-linear relationships between static demographics and dynamic behaviors. The key contributions of this work are as follows:

- We propose a model architecture based on MLP and LSTM to independently learn static and temporal features by decoupling each modal, thereby reducing feature interference and improving robustness.

- We introduce an intermediate fusion mechanism with a non-linear Aggregation block to effectively integrate multimodal features and capture complex relationships between modals beyond simple concatenation.
- We enable early prediction of student academic performance, allowing reliable identification of at-risk students using limited early stage data.

2 Related work

Early prediction of student academic performance remains a fundamental challenge in the field of learning analytics. Numerous approaches have been explored to address this issue, with a strong emphasis on integrating heterogeneous data sources and applying advanced machine learning and deep learning techniques to enhance predictive accuracy. This section provides an overview of the most prominent methods and strategies proposed in recent literature to tackle this problem.

Chango et al. [16] explored the integration of multi data sources to predict student academic performance. Their study applied several data fusion strategies, including merging all features, selecting optimal subsets of attributes, and employing ensemble learning techniques. Six classification algorithms (Jrip, Nnge, PART, REPTree, and Random Tree) were evaluated with the highest accuracy reaching 83.29%. Nevertheless, the study was limited by a small dataset comprising only 57 students from a single course, which restricts the generalizability of the findings. Moreover, the predictions were generated only after course completion, thereby limiting their usefulness for timely educational interventions.

To address the limitations of traditional machine learning approaches, Al-Azazi et al. [17] utilized deep learning models (RNNs, LSTMs, and GRUs) to transform OULAD into a time-series format and predict student outcomes over time. Their models achieved up to 95% accuracy, outperforming conventional approaches. The study treated both static (e.g., demographic) and dynamic (e.g., interaction) data uniformly by feeding them directly into the LSTM model. This introduced bias in feature learning and reduced the effectiveness of multimodal data integration.

In addition, Al-Zawqari et al. [18] proposed an automated feature extraction framework that eliminated the need for manual feature engineering. The model achieved 93% accuracy in predicting student dropouts using OULAD. But it struggled to identify exceptional students, with only 69% precision and 57% recall for the “Distinction” class. This shortcoming was primarily attributed to class imbalance in the dataset.

To overcome this limitation, Shou et al. [19] introduced a hybrid architecture that combines LSTM, artificial neural networks (ANN), and a multi-head attention mechanism to analyze multidimensional time-series data from OULAD. This model demonstrated strong performance, with F1-scores ranging from 73 to 99%, and provided valuable insights for early identification of at-risk students. Despite its effectiveness, the model's complex architecture made it prone to overfitting, and the absence of defined training protocols across different temporal granularities (e.g., daily vs. weekly inputs) limited its practical deployment.

Similarly, the hybrid approach using MLP and LSTM proposed by Karimi et al. [25] adopts a cumulative temporal modeling strategy in which each LSTM block processes information up to a specific week and incrementally aggregates features over time where the output of n^{th} week is calculated by $LSTM(week_1 + \dots + week_n)$. While the approach

works reasonably well, its cumulative design reduces the LSTM's ability to learn long sequences because each block sees only partial, aggregated inputs. To extend this idea, Prabowo et al. [20] proposed a multimodal deep learning model that integrates MLP and LSTM components to predict students' GPA. In this approach, the MLP processes structured features such as assessment scores and demographics, while the LSTM handles temporal data. The model achieved promising results, with a MSE of 0.4142 and a MAE of 0.418 on a 4.0 GPA scale. However, the model jointly processes all modalities from the beginning and continuously concatenates each layer, which constrains its ability to learn modality specific representations. This approach increases its sensitivity to noise, particularly when handling heterogeneous or variable length time series data. Furthermore, its reliance on simple feature concatenation limits the capacity to model complex, nonlinear interactions across modalities, thereby reducing overall robustness and generalizability.

From this review, it is evident that while machine learning and deep learning approaches have achieved notable success in academic performance prediction, several critical challenges remain. Existing methods often employ simplistic data fusion strategies that fail to capture the heterogeneity of educational data. Many models struggle to maintain performance across different learning environments or to differentiate effectively between high performing student subgroups. Furthermore, deep learning models frequently encounter overfitting due to overly complex architectures and the lack of uniform training procedures.

3 Methodology

3.1 Datasets

To evaluate the proposed model and compare it with existing methods, we use the OULAD dataset [26] and a self-collected dataset to assess the results.

3.1.1 OULAD dataset

The OULAD dataset includes assessment scores, demographic data, and clickstream interactions collected from an online learning environment. This integration enables efficient development of machine learning models for analyzing student behavior over time. OULAD contains data from 22 course offerings across seven distinct subjects, each spanning multiple semesters. Semesters are identified by "B" (start in February) or "J" (start in October) during 2013 and 2014.

The dataset is structured into three data sources. First, Demographic Information consists of details such as student ID, field of study, gender, place of birth, GPA, registration history, and any special requirements. Second, Assessment Information includes each student's grades and is classified into three categories: TMA (Tutor Marked Assessment), CMA (Computer Marked Assessment), and Final exam. Finally, Online Interaction Data is summarized from log files of the Moodle online learning platform, providing aggregated statistics on the total number of clicks each student made across different course elements.

The OULAD dataset comprises seven distinct courses, each with a varying number of participating students. This situation may lead to some students enrolling in multiple courses simultaneously, resulting in data overlap and affecting the consistency of learning behavior analysis. To address this issue, our study focuses on an in-depth analysis

of a single course, CCC, in order to ensure consistency in the behavioral data for each student.

3.1.2 Self-collected dataset

This dataset consists of four main sources of data: Demographic data, learning style survey data, online interaction data, and component grades. Notably, unlike the OULAD dataset, our self-collected dataset additionally includes learning style survey data, which is used as an assessment method to help instructors better understand learners. The VARK survey [12] reflects each student's learning style, represented by four values V, A, R, and K, each corresponding to one of four learning styles. Each value ranges from 0 to 16, as illustrated in Fig. 1.

To ensure privacy and data protection, all collected data were anonymized by removing personally identifiable information, including student IDs, phone numbers, and email addresses, and replacing them with automatically generated identifiers. All participants were fully informed of the study objectives and the intended research use of the data, and their participation was entirely voluntary. The collected data were used solely for academic research purposes and not for any commercial activities. In accordance with cultural norms and in compliance with Law No. 91/2025/QH15 on Personal Data Protection, which was enacted by the National Assembly of Vietnam on June 26, 2025, this study was exempt from formal ethical approval. The research did not involve the collection, storage, or processing of personally identifiable information, nor did it entail any intervention or manipulation of human participants. Furthermore, the study posed no identifiable risks to the participants. Ethical compliance was verified and confirmed by the Academy of Construction Strategy and Cadres Training, under Certificate No. 08/ACST-KHCN, which formally recognized the study's exemption from a comprehensive ethical review process.

Our self-collected dataset was systematically organized into Excel files, with each file representing a distinct type of information pertaining to students enrolled in the same course. Data were collected from cohorts 72 and 73 of the Introduction to Computer Science module during the 2023–2024 academic year at Hanoi National University of Education (HNUE), Viet Nam, as part of a doctoral research program hosted by the Academy of Construction Strategy and Cadres Training. All participants were undergraduate students belonging to STEM faculties, thereby ensuring a relatively homogeneous academic background. To construct the dataset, raw information was extracted

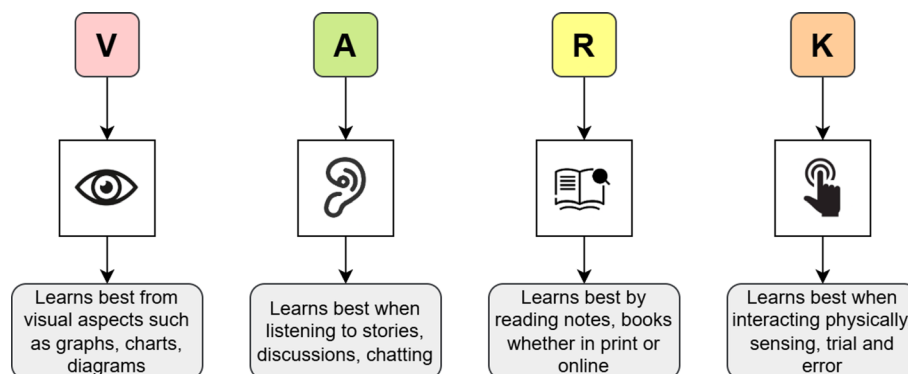


Fig. 1 The VARK learning style survey

from log files generated by the university's Moodle based learning management system (LMS), accessible at *cst.hnue.edu.vn*. Each log record was linked to a unique student identifier, which was subsequently anonymized and encrypted to guarantee data security and preserve participant privacy. This procedure enabled precise tracking of student activities while maintaining ethical standards for data handling.

In addition to system generated interaction logs, supplementary data were obtained through surveys. Specifically, a learning style survey was administered in parallel with a post course satisfaction survey conducted at the conclusion of the module. For the purposes of this study, we only focused exclusively on the learning style survey data for subsequent analysis. Table 1 below provides a detailed description of the dataset.

3.2 Data processing

For the OULAD dataset, we focus on processing data belonging to the CCC course. We normalize and aggregate online interaction data with component grades each week based on the course start date, resulting in a 9-week time series with 1,043 samples. During preprocessing, the Index of Multiple Deprivation band (*imd_band*) contained a small number of missing values. For these cases, we imputed the missing *imd_band* by assigning the value from other records belonging to the same region. This approach maintains regional consistency while avoiding the use of external information or the introduction of label dependent bias.

The proposed model then uses two input modalities: tabular demographic attributes and weekly time series capturing learning behaviors and assessment scores from Week 1 to Week 8, while the final grade in Week 9 is used as the prediction target. After applying categorical encoding [27] and z-score normalization (Formula 1), the processed data from 1043 students are fed into the deep learning models for feature extraction.

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

where x is the data point, μ is the mean and σ is the standard deviation of the dataset.

During log processing on our self-collected dataset, we observed that many recorded actions mainly represented user identification or system states rather than meaningful learning behaviors such as “File updated”, “Remove submission confirmation viewed”, etc. These actions were therefore manually removed and excluded from the analysis. We then performed a correlation analysis among the remaining action types to eliminate

Table 1 The self-collected dataset description

ID	Data sources	Column name
1	Demographics	Student ID, Department, Gender, Age, City, Past experience, Registration purpose (anonymized and encrypted)
2	Online interactions	Course viewed, Course module viewed, Quiz attempt viewed, Course activity completion updated, Sco launched, Quiz attempt reviewed, Quiz attempt submitted, User graded, Quiz attempt summary viewed, The status of the submission has been viewed, Submission submitted, File submission created, File uploaded, Submission form viewed, Quiz attempted, SCORM submitted, File uploaded, Submission status updated, Remove submission confirmation viewed, Step shown, Tour started, Gradebook user report viewed, Tour ended, Tour reset, User report viewed, Users list viewed, Profile viewed, Gradebook overview report viewed, Grade item updated, Comment added, Course searched
3	Assessment scores	Component scores, Final exam score
4	VARK learning style	Visual (V), Auditory (A), Read/Write (R), Kinesthetic (K)

highly correlated ones as shown in Fig. 2. All filtering and correlation analyses were performed exclusively on the training set to prevent potential information leakage. Despite this step, the number of action types was still relatively large, consisting of 20 various actions. As a result, we performed an additional step to identify the most important action types for analysis by using feature importance calculation.

Feature importance is an indicator that evaluates the influence of a feature on a machine learning model [28]. A higher value indicates that the feature has a greater impact on the model's prediction results. In this case, we applied a Random Forest model on the raw interaction dataset including 1,253 samples and 20 features using an 80–20 split on training and testing, then calculates the Gini index for each feature according to

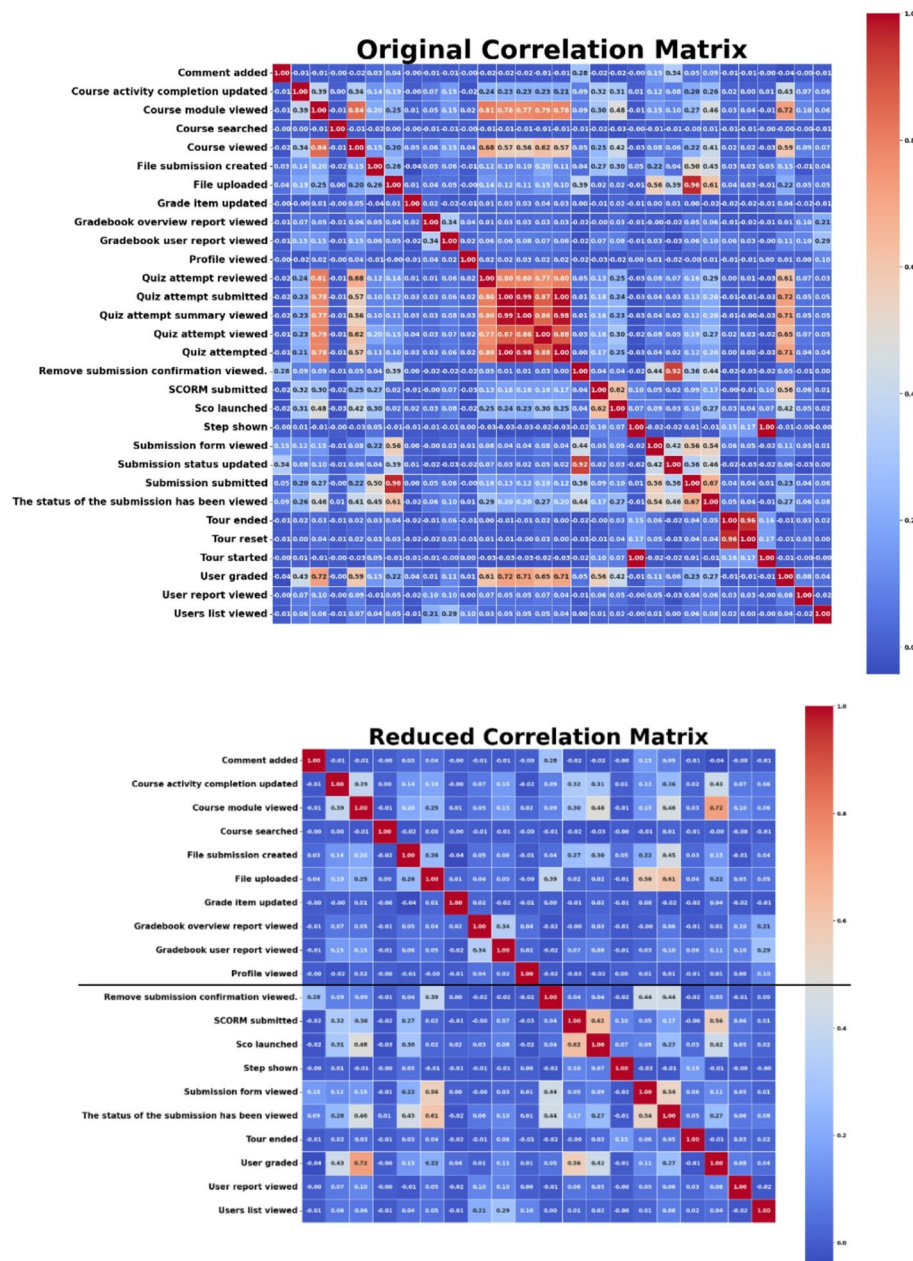


Fig. 2 Correlation analysis between types of online learning interactions. **a** The correlation matrix of original data. **b** The correlation matrix after removing highly correlated actions (*threshold* = 0.8)

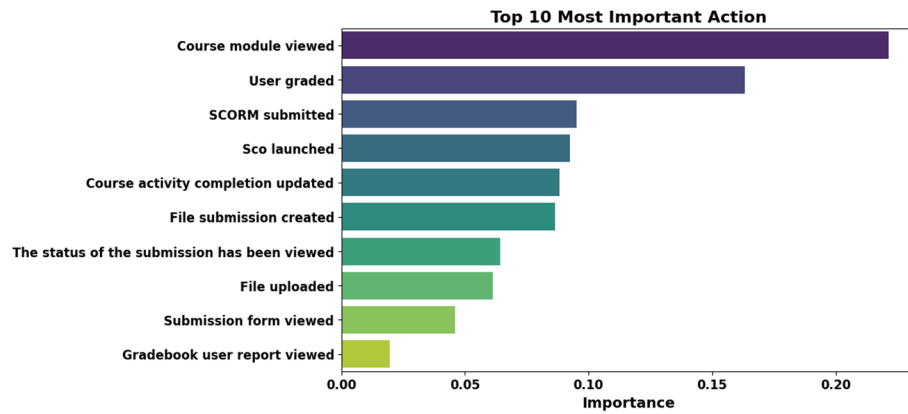


Fig. 3 Top 10 most important interactions

Table 2 The reference Table for 10-Point and 4-Point Grading Scales

ID	10-Point grade	4-Point grade
1	$8.5 < x < 10$	4
2	$7.8 < x < 8.4$	3.5
3	$7.0 < x < 7.7$	3.0
4	$6.3 < x < 6.9$	2.5
5	$5.5 < x < 6.2$	2.0
6	$4.8 < x < 5.4$	1.5
7	$4.0 < x < 4.7$	1.0
8	$x < 4.0$	0

Formula 2, and ranks the features in descending order of importance. To ensure a fair and evaluation, feature importance was computed using only the training data.

$$\text{Gini} = 1 - \sum_{i=1}^C (\{p_i\})^2 \quad (2)$$

where C is the number of classification classes; $p_i = \frac{n_i}{N}$, is the number of samples in i_{class} and N is the total number of samples. The Gini index, introduced in the CART decision tree algorithm, measures the purity of the data at each node in the tree. A higher Gini value indicates that the data at the node is purer, meaning the corresponding feature plays a more important role in the model. Based on these scores, the top 10 most informative action types were selected, illustrated in Fig. 3.

Each student's weekly online interaction score was then calculated as the sum of the frequencies of these 10 actions. The final processed dataset thus represents weekly student activity using the aggregated counts of the selected action types. This entire pre-processing pipeline was applied exclusively to the self-collected dataset and not to the OULAD dataset, in order to maintain the objectivity of the public benchmark.

Similar to the OULAD dataset, online interaction data from our self-collected dataset is divided into 10 weeks. Data from Week 1 to Week 9 is used to train the model, while the final exam score in Week 10 serves as the prediction label. After preprocessing, this dataset contains information from 793 students. Additionally, component grades in this dataset are also collected weekly and normalized from a 10-point scale to a 4-point scale to ensure compatibility with the GPA system in the OULAD dataset, as illustrated in Table 2.

3.3 Proposed framework

This study aims to predict students' academic performance at an early stage by leveraging both static and dynamic weekly data, enabling timely interventions and the delivery of personalized learning experiences to enhance student outcomes. We utilize multi-modal data comprising demographic information, assessment scores, online interactions, and learning style profiles, drawn from the publicly available OULAD dataset and an additional self-collected dataset. We propose a hybrid deep learning framework that combines a MLP and a LSTM network with separate feature extraction branches and an aggregation block to effectively model multimodal data. The MLP is employed to capture static attributes such as demographic features, while the LSTM is used to extract temporal patterns from sequential data, including weekly assessments and interaction logs.

Building on prior work [19, 20], our main contributions are summarized as follows:

- We propose a flexible LSTM architecture with a fixed maximum sequence length covering the full study period, allowing adaptation to varying weekly inputs while preserving consistency across samples. Unlike previous approaches, our model fuses modalities after feature extraction using an intermediate fusion strategy, enabling richer representations of static and temporal features and improving prediction accuracy.
- The proposed model employs specific feature extraction branches where each branch processes a distinct data source before the fusion stage. This design enables independent representation learning within each branch, thereby reducing interference during training. In particular, when dealing with noisy and temporally inconsistent data, noise from any branch does not affect others, allowing the remaining branches to remain stable and robust.
- An Aggregation block is introduced to transform sparse features from earlier layers into dense representations, enabling the model to capture complex and nonlinear relationships that simple feature concatenation cannot provide.

Overall, the proposed model comprises three components: (1) a feature extraction module, (2) a feature aggregation block, and (3) a prediction layer. The input consists of three parallel data processing blocks, each handling a specific input modality from Week 1 to Week k (w_1 to w_2), followed by a feature fusion stage and fully connected layers for final prediction, as shown in Fig. 4.

3.3.1 Feature extraction

During feature extraction, we use MLP and LSTM blocks illustrated in Fig. 5 to learn hidden representations for each data type.

We use Formula (3–8) of the LSTM architecture to learn the representation of each week as follows:

$$f_t = \sigma_g(W_t x_t + U_f h_{t-1} + b_f) \quad (3)$$

$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i) \quad (4)$$

$$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o) \quad (5)$$

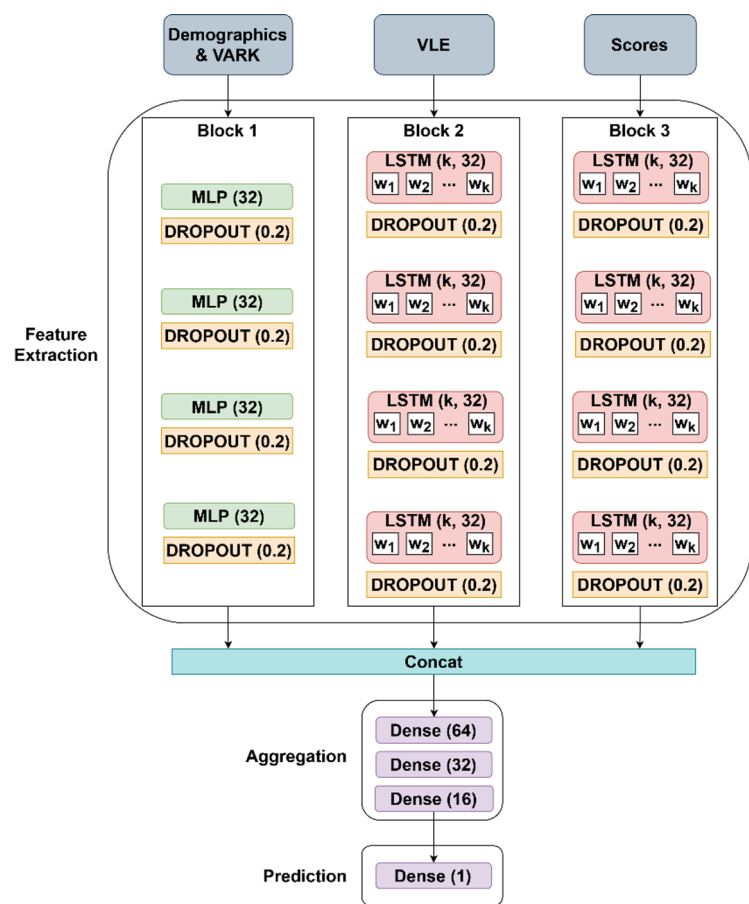


Fig. 4 Our proposed deep-learning based multimodal learning analytics model

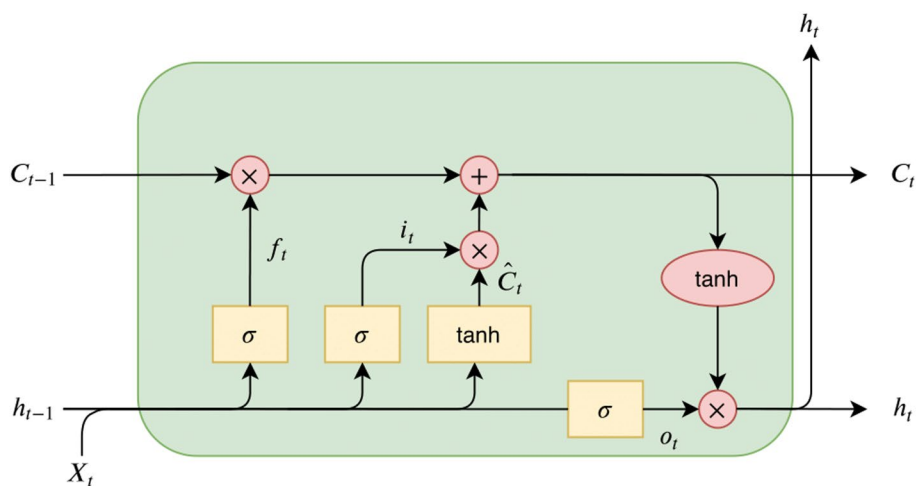


Fig. 5 LSTM overall architecture

$$\tilde{c}_t = \sigma_c (W_c x_t + U_c h_{t-1} + b_c) \quad (6)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (7)$$

$$h_t = o_t \odot \sigma_h (c_t) \quad (8)$$

where i_p , f_p , o_p , h_t and c_t respectively denote the input gate vector, forget gate vector, output gate vector, output unit vector, and memory cell vector. W , U and b are the parameters of the LSTM and a non-linear activation function Sigmoid $\sigma(\cdot)$.

Afterward, the extracted features are grouped based on their specific characteristics:

Block 1—Demographic and Learning Style Features

The first block processes static, non-sequential features such as demographic information and VARK learning style data. We use categorical encoding to convert student responses into numerical form and standardize the VARK survey data using z-score normalization to maintain the distribution of results for each survey question. Demographic factors such as age, gender, and experience give useful background about each student, while the VARK model shows how they prefer to learn. Both types of information are structured in tabular format, making them non-sequential in nature and suitable to combine within a unified processing framework. Using this information helps the model make more personalized and accurate predictions based on each student's unique traits and learning style. This data is then fed into a multi-layer perceptron model to extract non-linear representations. The MLP structure consists of four sequential hidden layers, each containing 32 nodes and using the ReLU activation function. To minimize overfitting, each layer is followed by a Dropout layer with a rate of 0.2. The process is represented by Formula (9–12):

$$h_1 = \text{ReLU}(W_1 x_{demo} + b_1), h_1 = \text{Dropout}(h_1, 0.2) \quad (9)$$

$$h_2 = \text{ReLU}(W_2 h_1 + b_2), h_2 = \text{Dropout}(h_2, 0.2) \quad (10)$$

$$h_3 = \text{ReLU}(W_3 h_2 + b_3), h_3 = \text{Dropout}(h_3, 0.2) \quad (11)$$

$$h_4 = \text{ReLU}(W_4 h_3 + b_4), h_4 = \text{Dropout}(h_4, 0.2) \quad (12)$$

where x_{demo} is the input feature vector, (W_p, W_2, W_3, W_4) and (b_p, b_2, b_3, b_4) are the weights and biases of each layer respectively. The output h_4 represents the abstract feature representation of demographic and VARK data and will be fed into the feature fusion stage along with outputs from other blocks.

Block 2—Online Interaction Features

The second block processes time series data representing students' learning behaviors collected from the LMS system. These sequential features, including sign in frequency, activity counts, and time spent on learning resources, are modeled using an LSTM network to capture their temporal representation. We calculated the total number of clicks in each week for every student and then arranged them in a sequential data format to serve as input for the LSTM model. The detailed example of these sequential features is illustrated in Table 3.

The input is represented as a tensor $X_{vle} \in R^{T \times d}$, where T denotes the number of time steps (8 weeks for OULAD, 9 weeks for our own dataset) and d is the number of features per step. Temporal dependencies are modeled using a stack of LSTM layers with

Table 3 Example of weekly aggregated VLE data for individual students in the OULAD dataset

Student ID	Week number	Sum click
Student 1	1	130
...	...	...
Student 1	8	645
...	...	...
Student N	1	248
...	...	...
Student N	8	692

Table 4 Example of weekly aggregated component scores for individual students in the self-collected dataset

Student ID	Week number	Component Score
Student 1	1	2.5
...	...	...
Student 1	8	7
...	...	...
Student N	1	9
...	...	...
Student N	8	9.5

32 hidden units per layer and a dropout rate of 0.2 to reduce overfitting. The process can be represented by Formula (13–16).

$$h_5 = LSTM_1(X_{vle}), h_5 = Dropout(h_5, 0.2) \quad (13)$$

$$h_6 = LSTM_2(h_5), h_6 = Dropout(h_6, 0.2) \quad (14)$$

$$h_7 = LSTM_3(h_6), h_7 = Dropout(h_7, 0.2) \quad (15)$$

$$h_8 = LSTM_4(h_7), h_8 = Dropout(h_8, 0.2) \quad (16)$$

where the output h_8 represents the cumulative learning behavior of each student throughout the course.

Block 3—Assessment Score Features

Students' grades during the course, including attendance, weekly quizzes, and assignment scores, also have temporal aspects similar to VLE interactions.

This block captures the pattern of each student's learning performance over time using three stacked LSTM layers, extracting features that reflect progress or stagnation. Weekly component scores are arranged chronologically and used as sequential inputs to the LSTM, with an illustrative example provided in Table 4. The input is represented as a time series tensor $X_{scores} \in R^{T \times d}$ and is processed as Formula (17–20).

$$h_9 = LSTM_5(X_{scores}), h_9 = Dropout(h_9, 0.2) \quad (17)$$

$$h_{10} = LSTM_6(h_9), h_{10} = Dropout(h_{10}, 0.2) \quad (18)$$

$$h_{11} = LSTM_7(h_{10}), h_{11} = Dropout(h_{11}, 0.2) \quad (19)$$

$$h_{12} = LSTM_8(h_{11}), h_{12} = Dropout(h_{12}, 0.2) \quad (20)$$

where the output h_{12} represents the overall learning performance of each student over time.

3.3.2 Feature aggregation

After extracting various features from the three data sources (Blocks 1, 2, and 3), the model proceeds to the feature fusion stage by concatenating the output vectors from all three blocks using Formula 21.

$$z = \text{Concat} (h_4^{\text{demo}}, h_8^{\text{vle}}, h_{12}^{\text{scores}}) \quad (21)$$

The combined feature vector z is then fed into an MLP comprising three sequential Dense layers with progressively decreasing units (64, 32, and 16), each using the ReLU activation function represented by Formula (22–24).

$$z_1 = \text{ReLU} (W_5 z + b_5) \quad (22)$$

$$z_2 = \text{ReLU} (W_6 z_1 + b_6) \quad (23)$$

$$z_3 = \text{ReLU} (W_7 z_2 + b_7) \quad (24)$$

3.3.3 Model prediction

Finally, the output layer consists of a single node with $\hat{y}$ as the final output value representing the predicted score using Formula 25.

$$\hat{y} = W_8 z_3 + b_8 \quad (25)$$

This design allows the model to consolidate all the extracted and aggregated features from multiple input modalities, including demographic, online learning behavior, and sequential performance scores into a single predictive value. By integrating information from these heterogeneous sources, the model captures both the static and dynamic aspects of student learning interactions, effectively modeling complex relationships and temporal dependencies.

4 Results and discussions

To evaluate the effectiveness of our proposed model, we conducted experiments on two datasets: OULAD and a self-collected dataset, as described in Sect. 3. Both datasets were divided into a 60% training set, 20% validation set, and 20% test set. Since the objective of the model is to predict each student's score, we used three common metrics illustrated in Formula (26–28) for regression tasks to assess model performance, including:

Mean Squared Error (MSE): Measures the average squared difference between predicted and actual values, reflecting the average deviation from the true values in terms of squared units with

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (26)$$

where n is the total number of samples, y_i is the $i_{\text{actual value}}$, $\hat{y}_i$ is the $i_{\text{predicted value}}$.

Mean Absolute Error (MAE): Measures the average absolute difference between predicted and actual values, reflecting the average deviation in a more intuitive and easily understandable way with

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (27)$$

where n is the total number of samples, y_i is the $i_{\text{actual value}}$, $\hat{y}_i$ is the $i_{\text{predicted value}}$.

R^2 -score: It represents the proportion of variance in the dependent variable that is explained by the model and measures how well the predicted values approximate the actual target values with

$$R^2 - \text{score} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (28)$$

where y_i denotes the true values, $\hat{y}_i$ are the predicted values, and $\bar{y}$ are the mean of the true values.

To ensure the reliability and robustness of the experimental results, each experiment was repeated ten times using fixed seeds from 1 to 10. For each evaluation setting, we report the mean values of MSE and MAE of ten runs along with their corresponding variances, which helps to reduce the influence of randomness in model initialization and data shuffling and provides a more stable evaluation of model performance. We also conducted experiments using other models, including RF and vanilla LSTM, to objectively compare the performance of our proposed model. Additionally, we included the MLP-LSTM model by Prabowo et al. [20] for a more comprehensive evaluation. In all experimental runs, the proposed model was trained with a batch size of 128 and optimized using the Adam optimizer with a fixed learning rate of 0.001. To maintain consistent training conditions across all models and datasets, early stopping was not applied, and all models were trained on 100 epochs.

4.1 Experiment results

4.1.1 Proposed model

In this study, we combine various data sources such as demographic data, assessment scores, online interaction behaviors, and learning style into a single feature table, serving as input to the prediction model. To analyze the impact of the amount of data sources on prediction results, we conducted two experimental scenarios: (1) using two or three data sources (demographics, component grades, and learning styles) and (2) using all four data sources, including online interaction data. Table 5 presents the results of our proposed model based on the two metrics, MSE and MAE. Bold values in the table indicate the best results.

These results show that adding more data sources clearly improves the model's prediction performance on both datasets. Specifically, for OULAD, using three data sources instead of two reduced the MSE from 0.2603 to 0.2404 and the MAE from 0.3506 to 0.3413. For our collected dataset, increasing from three to four data sources lowered the MSE from 0.7868 to 0.7410 and the MAE from 0.7346 to 0.6636. In terms of the

Table 5 Our proposed model results on OULAD and self-collected dataset

Dataset	Number of data sources	MSE	MAE	R^2 -score
OULAD	2	0.2603 ± 0.0385	0.3506 ± 0.0307	0.7586 ± 0.0286
OULAD	3	0.2404 ± 0.0275	0.3413 ± 0.0259	0.7836 ± 0.0578
Self-collected	3	0.7868 ± 0.0837	0.7346 ± 0.0498	0.2432 ± 0.0301
Self-collected	4	0.7410 ± 0.1166	0.6636 ± 0.0309	0.2893 ± 0.0961

R^2 -score, a consistent improvement is also observed as additional data sources are incorporated. For OULAD, the score increases from 0.7586 to 0.7836 when moving from two to three data sources, while for the self-collected dataset, the score from 0.2432 to 0.2893 after integrating an additional modality. This confirms that integrating more data sources, especially time-series data such as VLE, boosts prediction accuracy, as seen in the lower MSE values.

4.1.2 Vanilla LSTM

We also implemented a vanilla LSTM model with 32 hidden neurons and used the Adam optimizer to extend our experiments. The input data consists of weekly time-series sequences which are 8 weeks for the OULAD dataset and 9 weeks for our self-collected dataset. The model was trained for 100 epochs with a batch size of 64, aiming to predict each student's final score from the temporal features. We also tested the model's performance using different combinations of input data sources as before. Table 6 shows the results across two datasets, with varying input sources and number of features.

For the OULAD dataset, using two data sources including demographic information and assessment scores helps the LSTM model achieve an MSE of 1.0110 and an MAE of 0.8814. Adding online interaction data increased the number of features from 8 to 9, but the performance change was minimal where MSE slightly decreased to 1.0108, while MAE slightly increased to 0.8820. This shows that adding online interaction data made very small differences to the model's prediction performance on OULAD. In contrast, for the self-collected dataset, adding a fourth data source had a stronger effect. With three sources, the model had an MSE of 0.8726 and an MAE of 0.8169. After adding online interaction data, MAE dropped slightly to 0.8128, while MSE increased to 0.8784. Although the changes were small, the lower MAE suggests the model made slightly closer predictions in most cases. Overall, online interaction data can have some impact especially when it reflects actual learning behavior. However, since only one feature was added, the LSTM model in its current form does not show a significant difference when handling multidomain data.

4.1.3 Standard RF

For the standard RF model, we aggregate data weekly and combine it into a single feature table for each time step. Based on this structure, a separate RF model is trained for each week's data. This process is repeated every week: 8 weeks for the OULAD dataset and 9 weeks for our collected dataset. Much like the LSTM model, we also assess how the RF model performs with different combinations of input data sources. Performance results are shown in Table 7.

Using only demographics and component grades for OULAD, the model achieved an MSE of 0.3577 and an MAE of 0.3700. After adding online interaction data, the model performance improved noticeably with MSE dropping to 0.3060 and MAE to 0.3412.

Table 6 Experimental results of vanilla LSTM on OULAD and self-collected dataset

Dataset	Number of data sources	MSE	MAE
OULAD	2	1.0110	0.8814
OULAD	3	1.0108	0.8820
Self-collected	3	0.8726	0.8169
Self-collected	4	0.8784	0.8128

Table 7 Experimental results of standard RF model on OULAD and self-collected dataset

Dataset	Number of data sources	MSE	MAE
OULAD	2	0.3577	0.3700
OULAD	3	0.3060	0.3412
Self-collected	3	0.7435	0.6973
Self-collected	4	0.7458	0.6956

Table 8 Experimental results of MLP-LSTM model on OULAD and self-collected dataset

Dataset	Number of data sources	MSE	MAE	R ² -score
OULAD	2	0.2803 ± 0.0434	0.3864 ± 0.0250	0.7488 ± 0.0528
OULAD	3	0.2686 ± 0.0379	0.3836 ± 0.0282	0.7656 ± 0.0439
Self-collected	3	0.7796 ± 0.0340	0.7099 ± 0.0341	0.2161 ± 0.2078
Self-collected	4	0.7529 ± 0.1330	0.6898 ± 0.0570	0.2512 ± 0.1895

This shows that online learning behavior helps the RF model extract better features and improve prediction accuracy. On the other hand, for our self-collected dataset, adding the fourth data source had little impact. With three data sources, the model reached an MSE of 0.7435 and an MAE of 0.6973. After adding interaction data, MSE slightly increased to 0.7458 and MAE slightly decreased to 0.6956. This suggests that the interaction data in our dataset may lack the quality or relevance needed to help the model. In general, the standard RF model effectively uses online interaction data in the OULAD dataset, but its performance depends heavily on the quality and structure of the dataset.

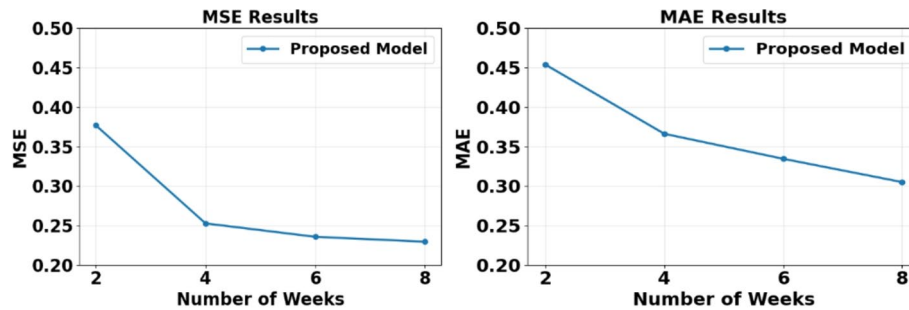
4.1.4 MLP-LSTM

We also investigated the MLP-LSTM model by Prabowo et al. [20], focusing on how its performance varied with the number of input data sources. Their original model predicted student scores based on GPA, drawing from demographic information and GPA data itself. To benchmark against our proposed model, we conducted two experimental setups: (1) we used the same two data sources as the original study and (2) we added a third data source which is online interaction data. We transformed the online interaction data into a time series similar to the GPA data by aggregating daily interactions and grouping them weekly for LSTM processing. The original MLP-LSTM architecture featured two distinct branches: one with 5 MLP layers for demographic data and another with 5 LSTM layers for GPA data. In our extended experiment, we added a third branch with 5 LSTM layers dedicated to handle the online interaction data. A key architectural detail is the vector addition used to link the MLP and LSTM layers within each branch. Across all layers, neuron counts were set to {32, 64, 128, 256, 512}, and Dropout rates ranged from {0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6}. The LSTM layers specifically employed the ReLU activation function and were optimized using Adam. Performance results are shown in Table 8.

When applied to the OULAD dataset, the MLP-LSTM model shows improved performance as additional data sources are incorporated. Specifically, increasing the number of input modalities from two to three reduces the MSE from 0.2803 to 0.2686 and the MAE from 0.3864 to 0.3836, while the R²-score increases from 0.7488 to 0.7656, indicating better predictive accuracy while using an additional data source. Similarly, on the self-collected dataset, adding a fourth data source leads to a clear improvement in performance, reducing the MSE from 0.7796 to 0.7529 and the MAE from 0.7099 to 0.6898,

Table 9 MSE and MAE results at different time step predictions

Number of weeks	MSE	MAE
2	0.3768	0.4539
4	0.2525	0.3662
6	0.2355	0.3345
8	0.2293	0.3050

**Fig. 6** MSE and MAE trends at different time step predictions

while also increasing the R^2 -score from 0.2161 to 0.2512. Overall, these findings indicate that integrating additional data sources consistently strengthens the MLP-LSTM model's predictive accuracy across both datasets, highlighting the importance of multimodal information in capturing diverse aspects of learners' behavior and performance.

4.1.5 Early prediction

To further validate the robustness of the proposed model, we conducted additional experiments by truncating the OULAD time series into shorter temporal segments and using these partial sequences to predict the final score. The corresponding results are summarized in Table 9 below.

Experiment results show that the predictive performance of the proposed model consistently improves as longer temporal histories are provided. When using only the first 2 weeks of data, the model achieves an MSE of 0.3768 and an MAE of 0.4539. Increasing the input length to 4 weeks substantially reduces the error to an MSE of 0.2525 and an MAE of 0.3662. Further gains are observed with 6 weeks of data, yielding an MSE of 0.2355 and an MAE of 0.3345. The best performance is obtained when 8 weeks of interaction data are used, with the MSE decreasing to 0.2293 and the MAE to 0.3050. The overall trends of MSE and MAE can be illustrated in Fig. 6.

These results demonstrate that the proposed model can produce reliable predictions even with limited early data. In the context of Multimodal Learning Analytics, this early predictive capability enables timely identification of students at risk of poor academic outcomes, thereby supporting proactive and targeted pedagogical interventions. As additional temporal information becomes available, the model further refines its predictions by leveraging longer learning histories, reflecting its ability to capture the evolving and multifaceted nature of student learning behavior.

4.2 Comparison between models

To evaluate the effectiveness of the proposed model, we compared its performance against three baseline models: RE, LSTM, and MLP-LSTM hybrid model, across two

Table 10 Comparison results of each model on OULAD

Model	2 data sources		3 data sources	
	MSE	MAE	MSE	MAE
RF (Table 3)	0.3577	0.3700	0.3060	0.3412
Vanilla LSTM (Table 4)	1.0110	0.8814	1.0108	0.8820
MLP-LSTM [20]	0.2803	0.3864	0.2686	0.3836
Our model	0.2603	0.3506	0.2404	0.3413

Table 11 Comparison results of each model on the self-collected dataset

Model	3 data sources		4 data sources	
	MSE	MAE	MSE	MAE
RF (Table 3)	0.7435	0.6973	0.7458	0.6956
Vanilla LSTM (Table 4)	0.8726	0.8169	0.8784	0.8128
MLP-LSTM [20]	0.7796	0.7099	0.7529	0.6898
Our Model	0.7868	0.7346	0.7410	0.6636

datasets (OULAD and a self-collected dataset). Evaluation was conducted using MSE and MAE as performance metrics.

The proposed model consistently outperformed the baseline methods on both datasets, achieving the lowest MSE and MAE values. On the OULAD dataset, with three data sources, our model achieved an MSE of 0.2293, compared to 0.2466 by MLP-LSTM, 0.3060 by RF, and 1.0108 by LSTM. Similarly, on our self-collected dataset, our model also outperformed others, achieving an MSE of 0.6161, compared to 0.7084 (MLP-LSTM), 0.7458 (RF), and 0.8784 (LSTM). Although the MLP-LSTM showed some benefit from the inclusion of interaction data in the OULAD dataset, its performance deteriorated on our dataset, suggesting limited generalizability. Tables 10 and 11 summarize the detailed MSE and MAE scores across different models and data configurations.

As Table 11 showed, on the OULAD dataset, our model achieved an MSE of 0.2603 and an MAE of 0.3506 with two data sources, improving further to 0.2404 MSE and 0.3413 MAE when a third data source was added. These results remain superior to RF, Vanilla LSTM, and MLP-LSTM, where performance gains from additional modalities were limited. A similar trend is observed on the self-collected dataset in Table 10. With three data sources, our model obtained an MSE of 0.7868 and an MAE of 0.7346, which improved to 0.7410 MSE and 0.6636 MAE when a fourth data source was incorporated. In contrast, the MLP-LSTM model exhibited small improvements or inconsistent behavior as more data sources were introduced. Overall, these findings demonstrate that the proposed architecture not only achieves better predictive accuracy but also scales more effectively with multimodal inputs, highlighting its robustness and suitability for multimodal learning analytics. Our experiment results can be summarized and illustrated by Figs. 7 and 8 below.

In early prediction scenarios, the proposed model consistently outperforms the MLP-LSTM [20] at all time steps. As shown in Table 12, it achieves lower MSE and MAE across all input lengths, including when only short temporal segments are available, demonstrating its effectiveness for early student performance prediction.

As the number of weeks increases, the error metrics steadily decrease on both models as shown in Fig. 9 indicates that longer historical context improves prediction performance. However, the proposed model benefits more from this additional context and maintains a clear performance advantage at every setting.

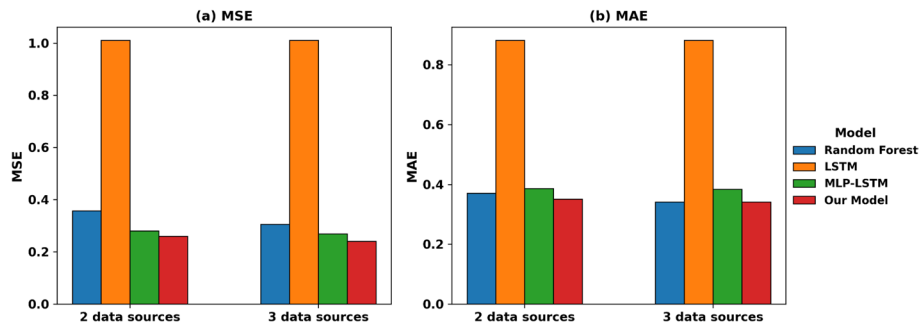


Fig. 7 Comparison results of each model on OULAD dataset

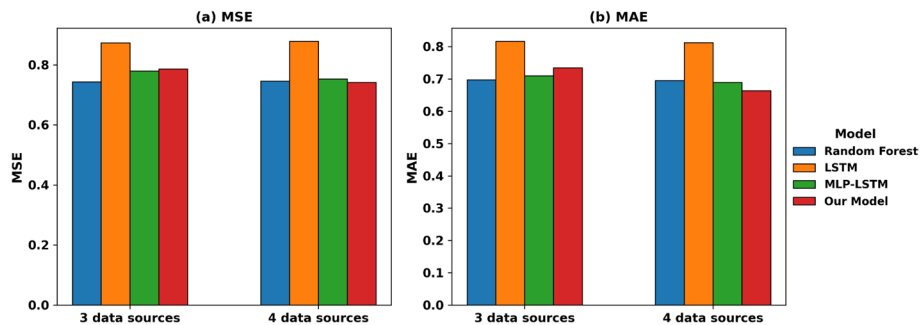


Fig. 8 Comparison results of each model on the self-collected dataset

Table 12 Comparison results between proposed model and MLP-LSTM [20] at different time step input

Number of weeks	MSE		MAE	
	Proposed	MLP-LSTM [20]	Proposed	MLP-LSTM [20]
2	0.3768	0.4379	0.4539	0.4792
4	0.2525	0.2649	0.3662	0.3713
6	0.2355	0.2530	0.3345	0.3648
8	0.2293	0.2466	0.3050	0.3511

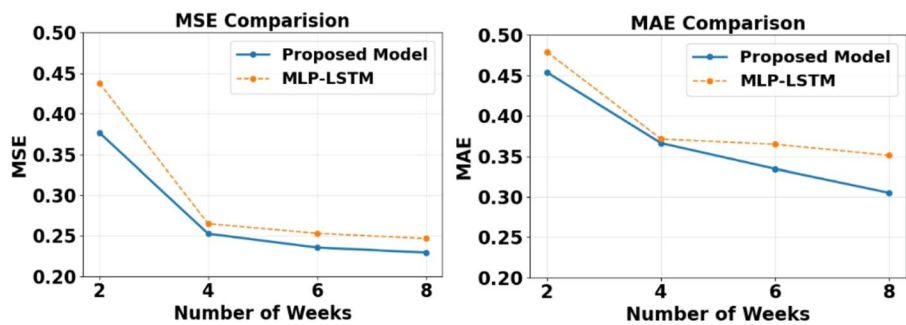


Fig. 9 Model trends comparison at different time step predictions

Table 13 Model parameters comparison

Parameters	Proposed model	MLP-LSTM [20]
Total parameters	103,745	4,214,177
Model size	405.25 KB	16.08 MB
Training time	~ 20 ms/epoch	~ 150 ms/epoch

Furthermore, we recorded and compared the number of model parameters in Table 13 while using identical computational resources, including an Intel i5-14600 K CPU, an NVIDIA 4070 Ti Super GPU with 16 GB of VRAM, and 32 GB of DDR5 RAM.

The proposed model contains substantially fewer parameters than the baseline MLP-LSTM [20], resulting in a much smaller model size and significantly faster training time per epoch. This efficiency gain is primarily attributed to the removal of inter-layer connections and the use of a smaller number of units in each layer, which lowers computational complexity without sacrificing learning capacity. Experimental results demonstrate that, despite its compact architecture, the proposed model consistently outperforms the MLP-LSTM [20] in terms of MSE, MAE, and R^2 -score, thereby confirming that a streamlined model design can achieve both improved predictive performance and practical efficiency.

Despite the encouraging results, several limitations of the proposed model should be acknowledged. The OULAD dataset and the self-collected dataset used in experiments was derived from a single course at one university, which restricts the diversity of learners and learning contexts represented in the data. As a result, the external validity and generalizability of the proposed model may be limited, as its performance could be influenced by specific course characteristics, assessment type, or institutional practices. Although the model demonstrates consistent improvements when integrating additional data sources, these findings may not directly transfer to other courses or educational settings with different learner behaviors and data distributions. Our future work therefore focuses on evaluating the proposed architecture on larger, multi-course and multi-institutional datasets to better assess its robustness and applicability across diverse educational environments.

5 Conclusions

This study proposed a deep learning based multimodal learning analytics model designed to predict student performance by effectively integrating different data sources, including demographic attributes, learning styles, online interaction logs, and weekly assessment score. By combining static and dynamic modalities, the proposed architecture captures complex relationships in student learning behavior while maintaining a compact and computationally efficient design. Experimental results on both the OULAD benchmark dataset and a self-collected dataset demonstrate that incorporating multiple data sources consistently improves predictive performance, achieving lower MSE and MAE and higher R^2 -scores compared with baseline models.

Beyond predictive accuracy, the proposed model provides a more holistic view of student learning processes, enabling the early identification of at-risk students and supporting timely educational interventions. The fusion of behavioral logs and performance trajectories with learner characteristics allows the model to uncover latent patterns in engagement and progression that are not observable from single-source data alone. These insights can assist educators in monitoring learning progress,

tailoring instructional strategies, and designing more personalized and adaptive learning experiences.

However, the generalizability of the findings is constrained by both the self-collected and OULAD datasets, as each dataset reflects a specific course and a specific learner population. While the model demonstrates strong performance within these settings, its applicability to other courses, disciplines, and educational environments requires further validation. Our future work will focus on extending the evaluation to more diverse educational contexts and exploring additional modalities to further enhance the robustness and applicability of multimodal learning analytics models in real world learning environments.

Author contributions

U.T.V and D.H.T proposed the idea and the model. U.T.V and B.H.T collected and processed the data and conducted the experiments. U.T.V drafted the manuscript. All authors read, revised, and approved the final version.

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Data availability

The datasets employed in this study are accessible at the following address: https://github.com/TranUoc/Dataset_MLA. These datasets are shared for academic and non-commercial research purposes only.

Declarations

Ethics approval and consent to participate

In compliance with Law No. 91/2025/QH15 on Personal Data Protection, which was enacted by the National Assembly of Vietnam on June 26, 2025, this study was exempt from formal ethical approval. It did not collect, store, or process personally identifiable information, nor involve any interventions or risks to participants. Ethical compliance was confirmed by the Academy of Construction Strategy and Cadres Training under Certificate No. 08/ACST-KHCN.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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